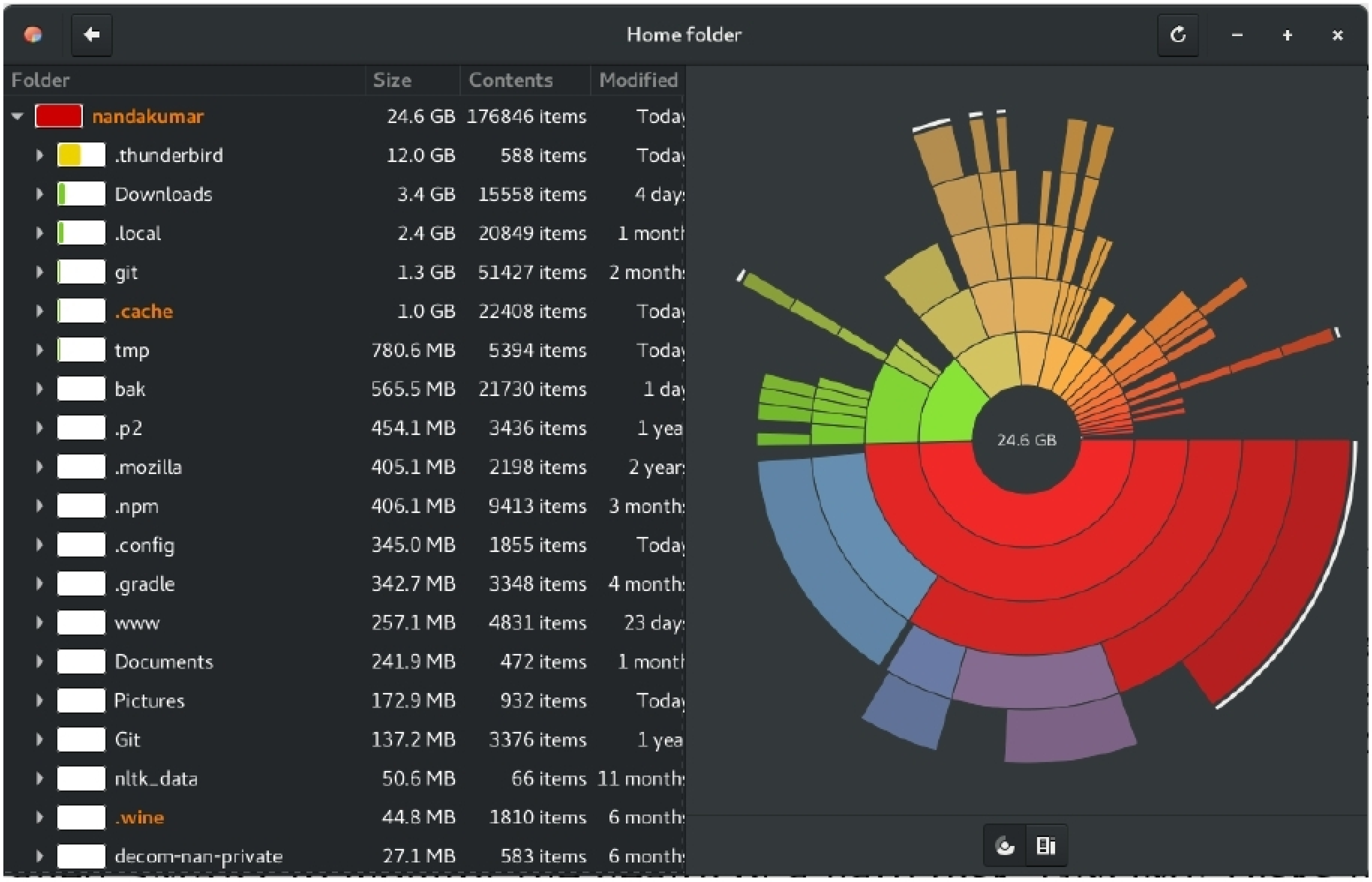

Figure 2: GNOME disk usage analyser (Baobab) visualises the space consumption of each directory

Disk partitioning and monitoring

Popular GNU/Linux distros come with some handy disk management tools like GNOME Disks which can do all the basic tasks including mounting, unmounting, partitioning, formatting and benchmarking. However, if you are into advanced stuff, it may be better to go with the tools made for them.

gparted is perhaps the most popular GUI partitioning tool used in GNU/Linux. Apart from partitioning, it offers resizing and file rescue. *gparted.org* offers a live CD version also. The power of *gparted* comes from *libparted*, which in turn is part of GNU Parted, a console utility for partition management.

More low-level utilities include *fdisk* and *gdisk*. Although the man pages of both utilities claim they understand GPT and MBR schemes, *fdisk* is recommended for MBR based partitioning and *gdisk* is recommended for GPT partitioning.

⚠ **Cautionary note:** There are multiple partitioning schemes. Those that were MBR based used to be the popular ones for a long time, but GPT is the current trend. Never perform partitioning operations until you understand the difference and find out which one you are currently using, especially if you are in fear of data loss or OS loss.

Partitioning just marks a region. Formatting, or assigning a file system type (e.g., Ext4, FAT32, NTFS, etc) is a must to make it useful. GUI partitioning tools incorporate this capability, while in the command line, you have to invoke separate tools like *mkfs*.

Disk monitoring: Monitoring the usage and health of disks is very important. Having no space left can sometimes cause apps to stop working and the OS to not boot, without any warning.

Total usage in partitions can be monitored using GNOME Disks, *gparted* or the command *df*. One can simply use the command *df -h* to know the free space of all mounted partitions (the *-h* option is for human-readable output).

Space consumption of directories can be monitored using the *du* command. *du -sh* will give you the total usage of the working directory (the directory your shell is in), while dropping the *s* option will cause the usage of each child to be listed. See the man page for more details. If you'd rather use a text-mode tool, use *ncdu*. Or if you are into GUIs, you can use a disk usage analyser (like Baobab).

There is a technology called SMART to monitor the health of a hard disk. GNOME Disks has a menu entry for this (one has to select the disk first). Alternatively, one can use the command *smartctl*, provided by the Debian package *smartmontools*. There is also